

The source boundary of extensional acyclic digraphs

An OEIS conjecture, finite-defect enumeration, and dyadic non-holonomicity

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In the mathematical research context of the ProveIt repository

Abstract

An extensional acyclic digraph has pairwise distinct out-neighborhoods. We give a self-contained proof of the formula $a(n) = n - \lceil \log_2 n \rceil$ for its maximum possible number of sources, the formula still marked conjectural in OEIS A182220 when consulted. The pigeonhole upper bound is already present in Tomescu's 2011 thesis; we credit that fact and do not present it as a new discovery.

The main development concerns the boundary of the source-count triangle A182162. If $u_m = A001192(m)$, $q = \lceil \log_2 n \rceil$, and $b_n^{(d)}$ counts isomorphism classes with exactly $n - q - d$ sources, we obtain an exact $(d + 1)$ -term formula and, in particular, $b_n^{(0)} = u_q \binom{2^q - q}{n - q}$. A uniform covering estimate then proves a dyadic entropy law for every fixed d . The complete set of subsequential exponential growth rates is an explicit interval, equal to $[1, 4]$ when $d = 0$. The exponential jumps across powers of two imply that none of these boundary sequences is P-recursive. Consequently their ordinary generating functions are not D-finite.

We also give an all-orders expansion of the Boolean-subset factor, retaining the small-core count exactly, and a total-variation binomial law for the number of missing arcs near a complete Boolean layer. The proofs are accompanied by independently implemented counting recurrences and exhaustive small-graph checks. These are conventional mathematical proofs with computational tests, not Lean-verified results. Historical priority for the boundary refinements is not established.

Primary sequences: A182220, A182162, A001192, A182161.

Keywords: transitive finite sets; source enumeration; Boolean layers; finite defects; entropy; polynomial recurrences; exact sampling.

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1 Problem, contribution, and prior-art boundary

1.1 What is being answered

Let $a(n)$ be the largest possible number of sources in an extensional acyclic digraph on $n \geq 1$ vertices. OEIS A182220 identifies this quantity and records, among other equivalent conjectural expressions,

$$a(n) = n - 1 - \lfloor \log_2(n - 1) \rfloor \quad (n \geq 3). \quad (1)$$

The entry attributes this formula to Antti Karttunen, August 17, 2013 [1]. For $n \geq 2$, the right side equals $n - \lceil \log_2 n \rceil$; the latter also gives $a(1) = 1$. The problem is not an issue of numerical fitting. We prove both the universal obstruction and an explicit construction for every admissible source count.

The more substantial question is: *how many graphs lie on, or a fixed distance below, this extremal boundary, and how do those counts behave?* Write $u(n, k)$ for the number of isomorphism classes with n vertices and k sources. The labeled count is $n!u(n, k)$; the factor $n!$ will be justified, not assumed. Define

$$q = q(n) = \lceil \log_2 n \rceil, \quad b_n^{(d)} = u(n, n - q - d), \quad d \in \mathbb{N}_0, \quad (2)$$

with value zero if the requested number of sources is not positive. For each fixed d , this convention affects only finitely many terms.

1.2 What was already known

Peddicord studied the numbers u_m of finite transitive, or full, sets [5, 3]. Policriti and Tomescu developed source-sensitive enumeration of extensional acyclic digraphs [6]. Tomescu's thesis explicitly gives the source bound $k \leq n - \lceil \log_2 n \rceil$ on printed page 29, along with source-counting recurrences and rigidity results [7]. The source sieve used below is also encoded in Johnston's 2012 Maple program in A182162 [2]. Wagner studied asymptotic enumeration and distributions in the unrestricted graph class [8].

Accordingly, neither the upper bound, the correspondence with full sets, rigidity, nor the general source sieve is claimed as new. An OEIS conjectural label does not establish that a statement is an unsolved research problem in the literature. Our proof supplies a direct resolution of the displayed formula; its upper-bound half has explicit prior art.

The exact boundary classification, the finite-defect specialization, the uniform entropy profile, and the non-P-recursiveness theorem form the development of this article. Some exact identities are short consequences of the classical source sieve. We make no assertion of historical priority for those consequences or for the probabilistic refinements. The meaningful claim is that complete proofs are supplied here, not that an exhaustive literature search has certified novelty.

1.3 Repository relationship and reproducibility

The supplied ProveIt repository was inspected at commit

6bf7f30d0352f7596e70928b3d4f304914075907.

Its organization separates mathematical sources, research reports, and executable certificates [9]. Searches for the present OEIS identifier did not locate an existing treatment. That limited search is not an exhaustive duplication audit. No repository theorem is used as an unverified premise below, and no repository files were modified.

The accompanying `verify.py` uses exact integers and independently implements a marked-source sieve and a source-deletion recurrence. It also enumerates all extensional DAG isomorphism classes through seven vertices. The finite tests support error detection; the proofs, rather than the tests, establish the infinite statements.

[write] **Added 3 October 2026, batch 85.** *Provenance.* This report is manuscript 07 of batch 85 of ProveIt’s docs/incoming intake: the archive `OEIS_A182220_Source_Boundary_Research.zip` (wrapper directory `oeis_source_boundary/`), arrival commit `9d6968c8a`, placed in `ddf8df5d5` (batch 85C), written against the pin `6bf7f30d0` quoted above. It is a single-source report. Its title block names no author and no tool. The write phase prefixed every label with `sbd:`, added the dated notes marked [write], and kept the title page out of the PDF page anchors; no statement, proof or number was changed.

Priority. The formula $a(n) = n - \lceil \log_2 n \rceil$ is *not* presented as a new discovery. Its upper half is Tomescu’s: on printed page 29 of the thesis [7], which the intake read, the count of unlabeled EADs with n vertices and s sources is said to vanish unless $2^{n-s} \geq n$, equivalently $s \leq n - \lceil \log_2 n \rceil$, “by extensionality and the pigeonhole principle”. What the present proof adds is the matching construction for every admissible source count (the sufficiency half of Theorem 3.1), itself elementary; no priority is claimed for it. The word “conjecture” in OEIS records the status of an entry’s formula in the encyclopedia. It does not certify that the problem was open in the literature, nor that the proof below is the first.

OEIS check. On 3 October 2026 the intake read A182220 on <https://oeis.org/A182220>. Its formula section lists four conjectural expressions: Karttunen’s (1) (2013), and others by W. I. Hurt (2014), P. Barry (2017) and A. Krivilev (August 2026, the “formula from August 2026” of the package’s `SOURCES.md`). Each is elementarily equivalent to (5) (Barry’s counts the powers of two in $[1, n - 1]$, Krivilev’s the non-powers of two there), and the intake confirmed all four against (5) for $n < 600$, as well as the 30 displayed terms. Theorem 3.1 therefore settles all four.

Repository. The claims of this subsection still hold at the time of writing: no other file of the repository mentions A182220, A182162, A001192, A182161, extensional acyclic digraphs, Tomescu, Policriti or Peddicord. No Lean or Rocq declaration formalizes any statement of this report, and its place in the research-report collection confers no formal status; Section 12.1 is a plan. Related material, none of it a shared theorem: (i) Lemma 2.1 is the classical Mostowski collapse, which the surreal report `Algebra/SurrealNumbers/docs/foundations-and-computation/birthday-cutoffs-and-hereditary-sets` proves for rooted well-founded codes as its `hset:lem:collapse`; (ii) the collection report `generating-functions-and-asymptotics/oeis-sequence-asymptotics/a116379-bounded-identity-trees` counts rooted identity trees, the other classical coding of hereditarily finite sets (it cites Meir, Moon and Mycielski); and (iii) Lemma 8.1 is a counterpart of a criterion of `a003407-dyadic-scaling-rigidity`, discussed in a note after it.

Notation. Several letters carry more than one local meaning:

- d is the defect, never the differential in d/dp or d/dz ; $B_d(z)$ in (33) is a generating function, while B_{2j} in Theorem 9.1 are Bernoulli numbers.
- $T = 2^m$ in Sections 7 and 9, but $T(n, k)$ in Appendix A is the labeled count $n! u(n, k)$ of A182162, as is the symbol $E(n, a(n))$ in (11), which the text does not define otherwise.
- C is a transitive core (and C , A constants in the proof of Lemma 8.1; $C_1(m, p)$ a correction coefficient); L is a source family, $L(z)$ the lacunary series of Proposition 3.2, and L also a truncation order in Theorem 9.1.
- $N = 2^m$ in the fixed-deletion and arc sections, N_r the jump indices of Lemma 8.1, $N(0, 1)$ the normal law; $p = n/T$ is a phase, p_j the recurrence polynomials of (32); R a removed family, R_d a radius; $M = 2^m - m$ and, in Theorem 6.1, $A = u_m \binom{M}{k}$.
- $\rho_n \in (1/2, 1]$ is the dyadic phase $n/2^q$, whereas $p = n/T$ with $T = 2^{q+d}$ lies in $(2^{-d-1}, 2^{-d}]$: the tempting reading “ $p = \rho_n$ ” is right only for $d = 0$.

No symbol was renamed.

2 Definitions and finite-set foundations

2.1 Orientation conventions

A digraph in this article is finite, loopless, and has no parallel arcs. An arc $v \rightarrow w$ means that w is an out-neighbor of v . A source has indegree zero; a sink has outdegree zero. A digraph is *extensional* when the map

$$v \mapsto N^+(v) = \{w : v \rightarrow w\}$$

is injective. It is acyclic when it contains no directed cycle. We abbreviate “extensional acyclic digraph” to EAD. Every nonempty EAD has exactly one sink: a DAG has a sink, and two sinks would have the same empty out-neighborhood. No transitive closure or transitive reduction of the arc relation is implicit in this definition.

Let u_m be the number of EAD isomorphism classes with m vertices, including $u_0 = 1$. We shall identify it with A001192. We use $\binom{M}{r} = 0$ if $r < 0$ or $r > M$ for nonnegative integer M . All logarithms without a subscript are natural.

2.2 Collapse, transitivity, and rigidity

Lemma 2.1 (Finite collapse). *For every EAD D , the recursion*

$$\phi(v) = \{\phi(w) : w \in N^+(v)\}$$

defines an injective map into the hereditarily finite sets. Its image $F = \{\phi(v) : v \in V(D)\}$ is a finite transitive set, and D is isomorphic to the membership digraph on F , with arcs $x \rightarrow y$ iff $y \in x$. Conversely, every finite transitive set yields an EAD this way. This gives a bijection between isomorphism classes of EADs and finite transitive sets.

Proof. Order the vertices so that every out-neighbor precedes its parent. The recursion is then finite and well defined. To prove injectivity, induct on the maximum height of two vertices, where height is the maximum length of a directed path from the vertex. Equal collapsed sets at height at most h have the same collapsed members. By the induction hypothesis the corresponding vertices of smaller height are identical. Thus their out-neighborhoods coincide, and extensionality identifies the two vertices. The base case is the unique sink.

Every member of $\phi(v)$ is another element of the image, so F is transitive: $\bigcup F \subseteq F$. Injectivity shows that membership reproduces exactly the arcs. Conversely, on a finite transitive set, Foundation precludes membership cycles, and Extensionality says that distinct elements have distinct member sets. These two constructions are inverse, including at the empty set. $\square$

Corollary 2.2 (Rigidity). *Every EAD has trivial automorphism group. Consequently the number of labeled EADs with n vertices and k sources is $n!u(n, k)$, and the total number is $n!u_n$.*

Proof. An automorphism preserves the recursive collapsed value of every vertex. The collapse is injective, so the automorphism fixes every vertex. There are therefore exactly $n!$ different labelings of each isomorphism class, and source count is preserved by relabeling. $\square$

This explains the normalization relating A182162 to $u(n, k)$ and A182161 to A001192 [2, 4, 3].

Lemma 2.3 (A sufficient bound on the small-core count). *For every $m \geq 0$,*

$$1 \leq u_m \leq 2^{\binom{m}{2}}. \tag{3}$$

Proof. The finite ordinal $m = \{0, 1, \dots, m-1\}$ is a transitive set of size m , so at least one class exists. Every DAG has a vertex ordering in which all arcs point to earlier vertices. For a fixed such ordering there are $2^{\binom{m}{2}}$ possible arc sets. Their isomorphism classes cover all DAG isomorphism classes, and hence all EAD classes. Injectivity of this covering is not needed for the upper bound. $\square$

3 The exact source maximum and all feasible source counts

Theorem 3.1 (Complete source support). *For $n \geq 1$ and $k \geq 1$, an EAD on n vertices with exactly k sources exists if and only if*

$$k \leq n, \quad n \leq 2^{n-k}. \quad (4)$$

In particular,

$$\boxed{a(n) = n - \lceil \log_2 n \rceil}. \quad (5)$$

Every integer from 1 through $a(n)$ occurs, so A182162 has no holes in the positive support of any row.

Proof. Suppose D has k sources and let C be its set of nonsources, $m = |C| = n - k$. No arc enters a source. Thus every out-neighborhood is a subset of C . The injective map $v \mapsto N^+(v)$ takes n vertices into a set of size 2^m , proving $n \leq 2^m$.

For sufficiency, first take $m = 0$. The conditions force $n = k = 1$, and the one-vertex edgeless digraph works. Suppose $m \geq 1$. Take a directed path on m core vertices, with arcs from each vertex to the preceding one. Its m out-neighborhoods are distinct. The full set C is not among them, since it would contain the vertex whose neighborhood it was, creating a loop. There are $2^m - m$ unused subsets of C , at least k by the assumed inequality. Choose k distinct unused subsets, one of them equal to C , and add a new vertex for each chosen subset, with exactly that subset as its out-neighborhood.

There are no arcs into the new vertices, so they are sources. All core vertices receive an arc from the new vertex whose neighborhood is C , so none is a source. Acyclicity and extensionality are immediate from the construction. Hence the graph has exactly k sources.

Finally, $n \leq 2^{n-k}$ is equivalent to $k \leq n - q(n)$. For $n \geq 1$, this upper endpoint is at least one. This proves both the maximum and the entire support assertion. $\square$

3.1 Elementary consequences for A182220

For $n \geq 2$, $\lceil \log_2 n \rceil = 1 + \lfloor \log_2(n-1) \rfloor$, which proves (1). Also,

$$a(n+1) - a(n) = 1 - \mathbf{1}_{\{n \text{ is a power of } 2\}}, \quad a(1) = 1. \quad (6)$$

This accounts for the repeated values. In particular the initial value 1 occurs three times, because both 1 and 2 are powers of two. For an interpretation by counting nonpowers of two, zero must be included:

$$a(n) = \#\{0 \leq j \leq n-1 : j \text{ is not a positive power of } 2\},$$

where “a power of 2” includes $2^0 = 1$. Omitting zero would create an off-by-one error. We do not assert every ancillary OEIS cross-reference without checking its conventions; the explicit formulas above remove that ambiguity.

3.2 A lacunary generating function for the maximum

The next consequence is included to separate the simple maximum sequence from the more intricate sequence counting its extremizers.

Proposition 3.2. *For $|z| < 1$,*

$$\sum_{n \geq 1} a(n)z^n = \frac{z}{(1-z)^2} - \frac{z}{1-z} \sum_{j \geq 0} z^{2^j}. \quad (7)$$

The unit circle is a natural boundary for this generating function.

Proof. Multiply the left side by $1-z$ and apply (6) to obtain the identity. Let $L(z) = \sum_{j \geq 0} z^{2^j}$ and let ζ be a root of unity of order 2^s . For all $j \geq s$, $\zeta^{2^j} = 1$; hence

$$L(r\zeta) = O(1) + \sum_{j \geq s} r^{2^j}, \quad r \uparrow 1,$$

and its real part tends to infinity. At every such $\zeta \neq 1$, the rational factor $z/(1-z)$ tends to a nonzero finite value, so the right side of (7) cannot extend holomorphically through ζ . These roots are dense on the circle. Analytic continuation across any point would give a neighborhood containing one of them, a contradiction. $\square$

The proof uses only a dense set of explicit singularities; no general lacunary-series theorem is needed.

4 Boolean-core classification and exact extremizer counts

4.1 The core is intrinsic

Let F be the collapsed transitive set of a graph. Its nonsources are exactly the elements which occur as members of other elements of F . Thus the nonsource core is

$$C = \bigcup F. \quad (8)$$

It is itself transitive. Indeed, $x \in y \in C$ means that $y \in z$ for some $z \in F$; then $y \in F$, and therefore $x \in \bigcup F = C$. If $L = F \setminus C$ is the set of sources, then

$$F = C \cup L, \quad L \subseteq \mathcal{P}(C) \setminus C. \quad (9)$$

The last exclusion uses transitivity of C , so that $C \subseteq \mathcal{P}(C)$. Conversely, (9) always defines a transitive set, but its nonsource core equals C only if $\bigcup(C \cup L) = C$. That covering condition is the sole extra constraint.

Lemma 4.1 (Automatic covering above half density). *Let C be any transitive set of size $m \geq 1$. If*

$$L \subseteq \mathcal{P}(C) \setminus C, \quad |L| = k, \quad m + k > 2^{m-1},$$

then $\bigcup(C \cup L) = C$.

Proof. If some $c \in C$ were outside the union, every member of the family $C \cup L$ would avoid c . All $m+k$ distinct subsets would then lie in $\mathcal{P}(C \setminus \{c\})$, which contains only 2^{m-1} subsets. $\square$

Theorem 4.2 (Complete classification of extremizers). *Let $n \geq 1$, $q = \lceil \log_2 n \rceil$, and $k = n - q$. The EAD isomorphism classes attaining $a(n)$ sources are in bijection with pairs*

$$C \in \mathcal{U}_q, \quad L \in \binom{\mathcal{P}(C) \setminus C}{k}, \quad (10)$$

where $\mathcal{U}_q$ is the set of transitive sets of size q . Consequently

$$\boxed{b_n^{(0)} = u_q \binom{2^q - q}{n - q}, \quad E(n, a(n)) = n! u_q \binom{2^q - q}{n - q}}. \quad (11)$$

Proof. Every extremizer has q nonsources by Theorem 3.1, and (8)–(9) recover its unique pair (C, L) . Conversely, if $q \geq 1$, then $n > 2^{q-1}$, so Lemma 4.1 shows that every pair in (10) has precisely C as its core and precisely L as its source set. No two different pairs give the same transitive set, since the core is its union. Each core offers exactly $2^q - q$ available subsets. Rigidity supplies the labeled factor. For $n = 1$, $q = 0$, and the single pair is $C = \emptyset$, $L = \{\emptyset\}$, giving the same formula. $\square$

Corollary 4.3 (Complete layers and sparse deletions). *For $m \geq 1$ and $0 \leq r < 2^{m-1}$,*

$$b_{2^m - r}^{(0)} = u_m \binom{2^m - m}{r}. \quad (12)$$

In particular $b_{2^m}^{(0)} = u_m$, and every extremizer on 2^m vertices is the membership graph of $\mathcal{P}(C)$ for a unique transitive m -element set C .

Proof. Complement the chosen source family in $\mathcal{P}(C) \setminus C$. The omitted family has size r , and at $r = 0$ no subsets are omitted. $\square$

Example 4.4. For $n = 8$ we have $q = 3$, $u_3 = 2$, and $b_8^{(0)} = 2$. For $n = 9$ the core size becomes four, and

$$b_9^{(0)} = u_4 \binom{12}{5} = 9 \cdot 792 = 7128.$$

The jump is not a numerical accident: the minimum allowed core size has increased, making a much larger Boolean layer available. Later sections quantify this phenomenon at every power of two.

5 The marked-source sieve and finite-defect diagonals

5.1 A classical identity, proved in the present normalization

Proposition 5.1 (Marked sources). *For $0 \leq t \leq n$,*

$$\sum_{s=t}^n \binom{s}{t} u(n, s) = u_{n-t} \binom{2^{n-t} - (n-t)}{t}. \quad (13)$$

For $n > 0$ we set $u(n, 0) = 0$; for $n = 0$ we set $u(0, 0) = 1$.

Proof. The left side counts an EAD isomorphism class together with a chosen t -element subset of its sources. Rigidity ensures that different chosen subsets cannot be identified by a nontrivial

automorphism. Delete the marked sources. Out-neighborhoods of the remaining vertices are unchanged, since no arc entered a source, so the remaining graph is an EAD on $n - t$ vertices.

Conversely, choose any such graph and choose t distinct subsets of its vertex set that are not already its out-neighborhoods. There are $2^{n-t} - (n - t)$ available subsets. Adjoin one marked source for each chosen subset. Existing sources are allowed to remain sources, which is why no covering constraint occurs in this construction. Deleting the marked sources reverses it. This proves the identity. $\square$

The identity is the marked-source version of the classical enumeration underlying the OEIS source sieve [2]. We use it as a proved starting point rather than a new counting principle.

Corollary 5.2 (Binomial inversion). *For $1 \leq k \leq n$,*

$$u(n, k) = \sum_{t=k}^n (-1)^{t-k} \binom{t}{k} u_{n-t} \binom{2^{n-t} - (n-t)}{t}. \quad (14)$$

Also

$$u_0 = 1, \quad u_n = \sum_{m=0}^{n-1} (-1)^{n-m-1} \binom{2^m - m}{n-m} u_m \quad (n \geq 1). \quad (15)$$

Proof. If $P_n(z) = \sum_s u(n, s) z^s$, then (13) is equivalent to

$$P_n(z) = \sum_{t=0}^n u_{n-t} \binom{2^{n-t} - (n-t)}{t} (z-1)^t.$$

Extract z^k to get (14). For $n > 0$, evaluate at $z = 0$, use $P_n(0) = 0$, and isolate the term $t = 0$ to get (15). $\square$

5.2 Only finitely many correction terms survive

Theorem 5.3 (Exact finite-defect formula). *Fix $n \geq 1$ and $d \geq 0$ with $k = n - q - d \geq 1$, and put $m = q + d$. Then*

$$b_n^{(d)} = \sum_{j=0}^d (-1)^j \binom{k+j}{j} u_{m-j} \binom{2^{m-j} - (m-j)}{k+j}. \quad (16)$$

Thus an entry at distance d below the source boundary requires exactly $d + 1$ displayed terms, using total counts only at the logarithmic indices $q, q + 1, \dots, q + d$.

Proof. Substitute $t = k + j$ in (14). A potentially nonzero term must satisfy

$$k + j \leq 2^{m-j} - (m - j), \quad \text{equivalently} \quad n \leq 2^{m-j}.$$

Since $q = \lceil \log_2 n \rceil$, this requires $m - j \geq q$, or $j \leq d$. The surviving terms are exactly those displayed. All have a positive binomial coefficient in the stated range, though signs alternate. $\square$

For $d = 0$, (16) recovers the exact extremizer count, so that count is also an immediate boundary specialization of the known sieve. At distance one it becomes

$$b_n^{(1)} = u_{q+1} \binom{2^{q+1} - q - 1}{n - q - 1} - (n - q) b_n^{(0)}. \quad (17)$$

At distance two, writing $k = n - q - 2$,

$$\begin{aligned} b_n^{(2)} &= u_{q+2} \binom{2^{q+2} - q - 2}{k} - (k+1)u_{q+1} \binom{2^{q+1} - q - 1}{k+1} \\ &\quad + \binom{k+2}{2} u_q \binom{2^q - q}{k+2}. \end{aligned} \quad (18)$$

These identities should not be mistaken for asymptotic expansions: they are exact integer formulas, even when early terms nearly cancel.

6 Uniform covering estimates and random cores

6.1 An estimate valid away from the extremal boundary as well

Theorem 6.1 (Uniform core-covering bound). *Let $m \geq 0$, $k \geq 1$, $n = m + k$, and $k \leq M = 2^m - m$. Set $A = u_m \binom{M}{k}$. Then*

$$0 \leq A - u(n, k) \leq m 2^{-k} A. \quad (19)$$

For $m \geq 1$, a sharper upper bound on the relative deficit is

$$\frac{A - u(n, k)}{A} \leq \frac{\binom{2^{m-1} - m}{k} \sum_s s u(m, s)}{\binom{2^m - m}{k} u_m}. \quad (20)$$

Proof. For a fixed transitive core C of size m , choose a uniform k -subset L of $\mathcal{P}(C) \setminus C$. The resulting graph is counted by $u(n, k)$ exactly when $\bigcup(C \cup L) = C$. Every element already in $\bigcup C$ is covered. If $c \in C \setminus \bigcup C$, then none of the m members of C contains c . Of the 2^{m-1} subsets avoiding c , precisely m are therefore forbidden core subsets. The probability that no chosen subset contains c is

$$\frac{\binom{2^{m-1} - m}{k}}{\binom{2^m - m}{k}} \leq 2^{-k}.$$

If the numerator vanishes the assertion is immediate. Otherwise compare successive factors of the two falling factorials; each ratio is at most $1/2$. Apply the union bound over the sources of C , whose number is at most m , and sum over all u_m cores. The sharper average-source version follows before replacing the number of sources by m . For $m = 0$, the only admissible choice is $k = 1$, and equality holds. $\square$

The average core-source count in (20) is itself exact: Proposition 5.1, with $t = 1$, gives

$$\frac{\sum_s s u(m, s)}{u_m} = \frac{(2^{m-1} - m + 1)u_{m-1}}{u_m} \quad (m \geq 1). \quad (21)$$

In applications where $k - \log_2 m \rightarrow +\infty$, the simpler bound already proves

$$u(m + k, k) = u_m \binom{2^m - m}{k} (1 + O(m2^{-k})). \quad (22)$$

The implicit error here is one-sided: the true value never exceeds the main expression.

6.2 Uniform sampling and the core law

There is an exact rejection sampler for a uniformly random isomorphism class with $n = m + k$ vertices and k sources. First draw a uniform core from $\mathcal{U}_m$, then a uniform k -subset of its M available neighborhoods. Keep the pair if it covers the core; otherwise repeat. Before rejection, every pair has exactly the same probability, and every accepted graph corresponds to one pair. Thus the accepted graph is exactly uniform. If $m2^{-k} < 1$, the expected number of trials is at most $(1 - m2^{-k})^{-1}$. This is conditional on having a procedure for uniformly sampling the much smaller core; it is not a claim of a complete polynomial-time sampler for arbitrary m .

At the extremal boundary, rejection is never necessary, and the core of a uniform extremizer is *exactly* uniform on $\mathcal{U}_q$. More generally, write δ for the actual rejection probability. Conditioning a finite probability distribution on an event of probability $1 - \delta$ changes it in total variation by exactly δ . Taking the core marginal cannot increase total variation. Therefore

$$d_{\text{TV}}(\text{core law of a uniform } (n, k) \text{ EAD}, \text{Unif}(\mathcal{U}_m)) \leq m2^{-k}. \quad (23)$$

This bound is uniform over all core structures. It explains why small-core statistics can be transferred to the high-source part of the triangle.

7 The dyadic entropy law

7.1 Uniform logarithmic asymptotics

Define the binary entropy, in natural units, by

$$H(p) = -p \log p - (1 - p) \log(1 - p), \quad H(0) = H(1) = 0.$$

For a fixed defect d , set

$$\rho_n = \frac{n}{2^{q(n)}} \in (1/2, 1], \quad \Phi_d(\rho) = \frac{2^d}{\rho} H\left(\frac{\rho}{2^d}\right). \quad (24)$$

The strict lower endpoint for ρ_n does not preclude subsequences converging to $1/2$.

Theorem 7.1 (Entropy profile). *For every fixed $d \geq 0$, as $n \rightarrow \infty$,*

$$\boxed{\log b_n^{(d)} = n\Phi_d(\rho_n) + O_d((\log n)^2)}. \quad (25)$$

The bound is uniform over the entire dyadic block, including its endpoints. No asymptotic formula for u_m is needed.

Proof. Let $m = q + d$, $T = 2^m$, and $k = n - m$. These parameters satisfy $m = O_d(\log n)$, $n \leq T < 2^{d+1}n$, and $k = n - O_d(\log n)$. Theorem 6.1 gives

$$\log b_n^{(d)} = \log u_m + \log \binom{T - m}{n - m} + O_d(m2^{-k}). \quad (26)$$

The small-core logarithm is $O_d((\log n)^2)$ by Lemma 2.3. The exact identity

$$\binom{T - m}{n - m} = \binom{T}{n} \frac{(n)_m}{(T)_m} \quad (27)$$

shows that the two binomial logarithms differ by $O_d(m)$: every factor $(n - i)/(T - i)$ is bounded below by a positive constant depending only on d , for all sufficiently large n .

For completeness, the elementary entropy bounds are

$$\frac{e^{TH(n/T)}}{T+1} \leq \binom{T}{n} \leq e^{TH(n/T)}. \quad (28)$$

For $0 < n < T$, consider a $\text{Bin}(T, n/T)$ random variable. Its probability at n is a maximum probability, hence is at least $1/(T+1)$ and at most one. Rearranging gives (28); the endpoint cases are immediate. It follows that $\log \binom{T}{n} = TH(n/T) + O(\log T)$. Finally $TH(n/T) = n\Phi_d(\rho_n)$. Combining these estimates proves (25), with no loss of uniformity as $n/T \rightarrow 1$. $\square$

7.2 Every subsequential exponential rate

Set

$$\lambda_d = 2^d H(2^{-d}), \quad d \geq 0. \quad (29)$$

In particular $\lambda_0 = 0$, $\lambda_1 = \log 4$, and $e^{\lambda_2} = 256/27$.

Theorem 7.2 (Complete growth-rate interval). *For each fixed $d \geq 0$, the set of subsequential limits of $(b_n^{(d)})^{1/n}$ is exactly*

$$[e^{\lambda_d}, e^{\lambda_{d+1}}]. \quad (30)$$

Thus

$$\liminf_{n \rightarrow \infty} (b_n^{(0)})^{1/n} = 1, \quad \limsup_{n \rightarrow \infty} (b_n^{(0)})^{1/n} = 4. \quad (31)$$

No fixed-defect boundary sequence has a single exponential growth constant.

Proof. Write $p = \rho/2^d$. Then $\Phi_d(\rho) = H(p)/p$. For $0 < p < 1$,

$$\frac{d}{dp} \frac{H(p)}{p} = \frac{pH'(p) - H(p)}{p^2} = \frac{\log(1-p)}{p^2} < 0.$$

Hence Φ_d is strictly decreasing, with values λ_{d+1} at $\rho = 1/2$ and λ_d at $\rho = 1$. Every $\rho \in (1/2, 1)$ is the limit of phases obtained by rounding $\rho 2^q$ to integers. Powers $n = 2^q$ realize phase one, while $n = 2^{q-1} + 1$ tends to phase $1/2$. The entropy theorem gives the entire claimed interval. Conversely, compactness of $[1/2, 1]$ and continuity of Φ_d exclude any other subsequential limits. $\square$

In particular the first three intervals, for $d = 0, 1, 2$, are

$$[1, 4], \quad [4, 256/27], \quad [256/27, 2^{24}/7^7].$$

The upper endpoint for one defect equals the lower endpoint for the next. This expresses the fact that crossing a dyadic boundary adds one core vertex, just as increasing the allowed source deficit does.

8 Non-P-recursiveness of every fixed-defect boundary

A sequence (c_n) is P-recursive if, for all sufficiently large n ,

$$\sum_{j=0}^h p_j(n) c_{n+j} = 0 \quad (32)$$

for polynomials p_j , not all zero. Allowing a finite exceptional initial segment makes the conclusion below stronger, not weaker. A formal power series is D-finite if it satisfies a nonzero homogeneous linear differential equation with polynomial coefficients.

Lemma 8.1 (An exponential-jump obstruction). *Suppose a nonnegative sequence (c_n) has integers $N_r \rightarrow \infty$ and constants $\alpha < \beta$ such that, for every fixed $j \geq 0$,*

$$c_{N_r-j} = \exp(\alpha N_r + o(N_r)), \quad c_{N_r+1} = \exp(\beta N_r + o(N_r)).$$

Then (c_n) is not P-recursive.

Proof. Suppose (32) holds, and choose h as the largest index with $p_h \neq 0$. If $h = 0$, positivity along (N_r) contradicts the recurrence. Otherwise apply it at $n = N_r + 1 - h$. For sufficiently large r , $p_h(n)$ is nonzero, and ratios of fixed polynomials give a bound

$$c_{N_r+1} \leq C N_r^A \sum_{j=0}^{h-1} c_{N_r+1-h+j} = \exp(\alpha N_r + o(N_r))$$

for constants C, A . This contradicts $\beta > \alpha$. Absolute values of the coefficients were used, so cancellation cannot invalidate the estimate. The argument also applies to complex polynomial coefficients. $\square$

[write] **Added 3 October 2026, batch 85.** The collection report `generating-functions-and-asymptotics/oeis-sequence-asymptotics/a003407-dyadic-scaling-rigidity` proves, in its Part II, an obstruction that runs the other way. Its Corollary `dsr:nh:cor:jumps` (a consequence of its local-growth rigidity theorem `dsr:nh:thm:rigidity`, which rests on a G-function input) says that an eventually positive, exponentially bounded, integer P-recursive sequence without an n th-root limit must have $u_{n+1}/u_n > e^{\varepsilon n}$ for infinitely many n : upward exponential jumps are *necessary* for such oscillation, and their absence rules P-recursiveness out. Lemma 8.1 is a *sufficient* condition in the opposite situation: a jump preceded by a plateau of every fixed length at the lower rate rules P-recursiveness out. It needs neither integrality nor an exponential bound, and its proof is elementary. The two criteria are complementary, not competing. The boundary sequences $b^{(d)}$ do have the upward jumps (from $e^{\lambda_d N_r}$ to $e^{\lambda_{d+1} N_r}$ at $N_r = 2^r$), so the a003407 criterion cannot exclude their P-recursiveness, and Lemma 8.1 is the one that applies. Conversely, the 3AP-free permutation counts of a003407 have only subexponential forward increases, so Lemma 8.1 does not apply to them. The plateau hypothesis “for every fixed $j \geq 0$ ” is essential: a003407’s example $3^n + (-3)^n + 2^n$, which satisfies a constant-coefficient recurrence, jumps from rate $\log 2$ at odd indices to $\log 3$ at even ones, but its two-term window never stays at the lower rate.

Theorem 8.2 (Non-holonomicity). *For every fixed $d \geq 0$, neither $(b_n^{(d)})$ nor $(n!b_n^{(d)})$ is P-recursive. The ordinary generating function*

$$B_d(z) = \sum_{n \geq 1} b_n^{(d)} z^n \tag{33}$$

is not D-finite. Its radius of convergence is

$$R_d = e^{-\lambda_{d+1}}; \quad R_0 = \frac{1}{4}. \tag{34}$$

Proof. Take $N_r = 2^r$. For every fixed $j \geq 0$, the index $N_r - j$ remains in the dyadic block with $q = r$ when r is large, and its phase tends to one. By Theorem 7.1,

$$\log b_{N_r-j}^{(d)} = \lambda_d N_r + o(N_r).$$

At $N_r + 1$, $q = r + 1$ and the phase tends to $1/2$, giving

$$\log b_{N_r+1}^{(d)} = \lambda_{d+1} N_r + o(N_r).$$

The strict decrease proved in Theorem 7.2 yields $\lambda_{d+1} > \lambda_d$. Lemma 8.1 applies.

If $n!b_n^{(d)}$ satisfied a polynomial recurrence, division by $n!$ would give one for $b_n^{(d)}$: each factor $(n+j)!/n!$ is a polynomial in n . Thus the labeled sequence is also non-P-recursive.

For completeness, extracting coefficients from a nonzero polynomial linear differential equation for B_d yields a nonzero finite-shift polynomial recurrence for its coefficients. Indeed, coefficients of $z^i(d/dz)^j B_d$ are shifted coefficients multiplied by falling factorials in the index; a common shift removes any negative indices. This contradicts what was just proved. Finally, the Cauchy–Hadamard formula and Theorem 7.2 give (34). $\square$

Remark 8.3 (What the theorem does not say). Non-D-finiteness does not by itself imply that the circle $|z| = R_d$ is a natural boundary, nor does it rule out useful nonlinear functional equations. We prove a natural boundary for the maximum sequence in Proposition 3.2, but do not make that stronger claim for B_d . The labeled ordinary generating series has radius zero; the series B_d is also the exponential generating series of the labeled boundary counts.

9 An all-orders expansion inside a dyadic block

The entropy law is uniform even at complete layers. More detailed prefactors require staying away from the degeneracy $p = 1$. This section retains u_m exactly; it does not purport to replace the classical, separate asymptotic analysis of the total EAD count.

Theorem 9.1 (Explicit expansion with exact core amplitude). *Fix $d \geq 0$ and $\varepsilon > 0$. Let*

$$m = q + d, \quad T = 2^m, \quad p = n/T, \quad k = n - m, \quad \varepsilon \leq p \leq 1 - \varepsilon.$$

Define $S_\ell(m) = \sum_{i=0}^{m-1} i^\ell$ and let B_{2j} denote the Bernoulli numbers with $B_2 = 1/6$. For fixed integers $L \geq 0$, $J \geq 1$,

$$\begin{aligned} \log b_n^{(d)} &= \log u_m + TH(p) - \frac{1}{2} \log(2\pi T p(1-p)) + m \log p \\ &\quad - \sum_{\ell=1}^L \frac{p^{-\ell} - 1}{\ell T^\ell} S_\ell(m) \\ &\quad + \sum_{j=1}^J \frac{B_{2j}}{2j(2j-1)T^{2j-1}} [1 - p^{1-2j} - (1-p)^{1-2j}] \\ &\quad + O_{d,\varepsilon,L,J} \left(\frac{m^{L+2}}{T^{L+1}} + T^{-(2J+1)} + m2^{-k} \right). \end{aligned} \quad (35)$$

The first sum is empty when $L = 0$.

Proof. Start with (26) and (27). The logarithm of the falling-factorial ratio is

$$m \log p + \sum_{i=0}^{m-1} \left[\log \left(1 - \frac{i}{pT} \right) - \log \left(1 - \frac{i}{T} \right) \right].$$

Since $m = O_d(\log T)$ and $p \geq \varepsilon$, expand both logarithms through degree L . The coefficient of $T^{-\ell}$ is exactly the first sum in (35). Summing the elementary Taylor remainders gives $O(m^{L+2}/T^{L+1})$.

Apply the Stirling expansion for $\log(T!)$, $\log((pT)!)$, and $\log(((1-p)T)!)$. Subtraction produces $TH(p)$, the displayed Gaussian prefactor, and the Bernoulli sum. Its remainder after J terms is $O_{\varepsilon,J}(T^{-(2J+1)})$. This standard expansion follows, for example, by Euler–Maclaurin summation of $\log x$; its remainder has the asserted bound because pT and $(1-p)T$ are both at least εT . Finally include the covering error from (26). $\square$

In particular,

$$b_n^{(d)} = \frac{u_m p^m e^{TH(p)}}{\sqrt{2\pi T p(1-p)}} \left[1 + \frac{C_1(m, p)}{T} + O_{d, \varepsilon} \left(\frac{m^4}{T^2} \right) \right], \quad (36)$$

$$C_1(m, p) = -\frac{1-p}{2p} m(m-1) + \frac{1-p^{-1} - (1-p)^{-1}}{12}.$$

Taking larger L and J supplies arbitrarily many logarithmic corrections. Exponentiating gives a multiplicative expansion whose coefficients are explicit polynomials in m , with rational functions of p as coefficients.

For $d \geq 1$, the actual phases have $p \in (2^{-d-1}, 2^{-d}]$, so a suitable fixed ε covers the entire dyadic block. For $d = 0$, the complete-layer edge $p \rightarrow 1$ must instead be treated by the exact deletion formula (12). This is a genuine boundary change of asymptotic regime, not an error to be removed by formally substituting $p = 1$ in the Gaussian prefactor.

9.1 The fixed-deletion regime

For fixed $r \geq 0$ and $N = 2^m$, Corollary 4.3 gives

$$b_{N-r}^{(0)} = u_m \frac{N^r}{r!} \left[1 - \frac{rm + r(r-1)/2}{N} + O_r \left(\frac{m^2}{N^2} \right) \right]. \quad (37)$$

This follows directly by expanding $\binom{N-m}{r} = N^r / r! \prod_{i=0}^{r-1} (1 - (m+i)/N)$. More generally, for $r = o(N)$ with $m+r < N$, an exact logarithmic series is

$$\log \frac{r! b_{N-r}^{(0)}}{u_m N^r} = - \sum_{\ell \geq 1} \frac{1}{\ell N^\ell} \sum_{i=0}^{r-1} (m+i)^\ell. \quad (38)$$

It converges because $(m+r-1)/N < 1$, and truncation is controlled by the geometric tail of the logarithm series. This supplies a separate exact route toward a matched sparse-to-bulk expansion.

10 Arc counts and a binomial law near complete layers

10.1 A core-dependent polynomial with a simple complement form

Let $e(F)$ be the number of arcs in the membership graph of F . Since arcs are membership incidences,

$$e(F) = \sum_{S \in F} |S|.$$

For $n = 2^m - r$ with $0 \leq r < 2^{m-1}$, an extremizer is $F = \mathcal{P}(C) \setminus R$, where $C \in \mathcal{U}_m$ and $R \subseteq \mathcal{P}(C) \setminus C$ has size r . Hence

$$e(F) = m2^{m-1} - Z, \quad Z = \sum_{S \in R} |S|. \quad (39)$$

In particular, *all* extremizers at $n = 2^m$ have exactly $m2^{m-1}$ arcs, regardless of the core graph.

For a fixed core C , the full deficit-counting polynomial is

$$[z^r] \prod_{S \in \mathcal{P}(C) \setminus C} (1 + zw^{|S|}). \quad (40)$$

Summing this over $C \in \mathcal{U}_m$ gives the exact unlabeled joint count by size and arc deficit. Equivalently, the product is the formal quotient

$$\frac{\prod_{j=0}^m (1 + zw^j)^{\binom{m}{j}}}{\prod_{x \in C} (1 + zw^{|x|})}.$$

The denominator divides the numerator as a polynomial product; it is not an analytic singularity assertion.

Theorem 10.1 (Total-variation binomial approximation). *Let F be uniform among extremal EAD isomorphism classes on $n = 2^m - r$ vertices, where $m \geq 1$ and $0 \leq r < 2^{m-1}$. Define $Z = m2^{m-1} - e(F)$. Then*

$$\boxed{d_{\text{TV}}(\mathcal{L}(Z), \text{Bin}(rm, 1/2)) \leq \min \left\{ 1, \frac{rm + \binom{r}{2}}{2^m} \right\}.} \quad (41)$$

The same statement holds for a uniformly labeled extremizer.

Proof. By Theorem 4.2, the core is uniform on $\mathcal{U}_m$, and, conditional on it, R is a uniform r -subset of the available family. Fix a core and draw r independent uniform subsets $S_1, \dots, S_r$ of the entire Boolean cube $\mathcal{P}(C)$. Each S_i has a $\text{Bin}(m, 1/2)$ size, independently; therefore $\sum_i |S_i|$ is exactly $\text{Bin}(rm, 1/2)$.

Condition this ordered sample on avoiding the m forbidden core subsets and on having no repetition. The conditional distribution is exactly uniform sampling without replacement from $\mathcal{P}(C) \setminus C$, with an irrelevant ordering. By the union bound, the conditioning event fails with probability at most

$$r \frac{m}{2^m} + \binom{r}{2} \frac{1}{2^m}.$$

Conditioning changes total variation by the failure probability, and mapping to the sum of the sizes cannot increase total variation. The estimate is independent of the core, so it survives mixing over cores. Rigidity implies that uniform labeling gives the same induced law on isomorphism classes, proving the last assertion. $\square$

Corollary 10.2 (Boundary central limit law). *If $m \rightarrow \infty$, $r = r(m) \geq 1$, and $r(m+r) = o(2^m)$, then*

$$\frac{m2^{m-1} - e(F) - rm/2}{\sqrt{rm}/2} \implies N(0, 1). \quad (42)$$

Proof. The total-variation error in (41) tends to zero. For $B \sim \text{Bin}(rm, 1/2)$, the characteristic function of $(B - rm/2)/(\sqrt{rm}/2)$ is $\cos(t/\sqrt{rm})^{rm}$, which tends to $e^{-t^2/2}$. The claimed convergence follows, since $rm \rightarrow \infty$. $\square$

For fixed $r \geq 1$, the error is $O_r(m2^{-m})$ and the Gaussian limit already applies. For $r = 0$, the distribution is instead deterministic. The theorem concerns graphs conditioned to maximize their source count; it is not Wagner's normal limit for unrestricted EADs.

10.2 Exact conditional moments

Let $N = 2^m$, $M = N - m$, and define two core statistics

$$e_C = \sum_{x \in C} |x|, \quad h_C = \sum_{x \in C} |x|^2.$$

The sum and squared sum of subset sizes in the available population are

$$A_1 = \frac{mN}{2} - e_C, \quad A_2 = \frac{(m^2 + m)N}{4} - h_C.$$

Writing $\mu_C = A_1/M$ and $v_C = A_2/M - \mu_C^2$, uniform sampling without replacement gives

$$\mathbb{E}(Z \mid C) = r\mu_C, \quad \text{Var}(Z \mid C) = \frac{r(M-r)}{M-1}v_C \quad (M > 1). \quad (43)$$

The variance identity follows by expanding the variance of the sum of r sampled population entries: the covariance of two distinct sampled entries is $-v_C/(M-1)$. This derivation also specifies the variance normalization. When $M = 1$, the sample is deterministic. These exact moments remain available outside the small-error range of Theorem 10.1.

11 Exact verification and computational artifacts

11.1 Three complementary checks

The accompanying program performs the following distinct checks. First, it computes u_n from (15) and then evaluates the marked-source sieve. Second, it computes the source triangle using the source-deletion recurrence from Tomescu's Corollary 2.1.7 [7]:

$$u(n, s) = \frac{1}{s} \left[(2^{n-s} - (n-1))u(n-1, s-1) + \sum_{j=0}^{n-s-1} \binom{s+j}{j+1} 2^{n-1-s-j} u(n-1, s+j) \right]. \quad (44)$$

Here $u(1, 1) = 1$ and out-of-range entries are zero. The code verifies exact divisibility by s , equality of the two triangle constructions, the complete support theorem, the finite-defect formulas, and the covering inequalities.

Third, for every $n \leq 7$, the program enumerates all graphs in which arcs point from a larger vertex index to a smaller one. It rejects repeated out-neighborhoods, collapses surviving graphs to nested finite sets, and deduplicates the resulting isomorphism classes. This does not use either counting recurrence. Source counts and arc counts are computed directly from the arc masks. Every DAG has such an ordering, so the enumeration covers every class, even though some classes occur under several orderings.

11.2 Recorded results

The recorded run used `--max-n 64 --brute-n 7`. It checked 2080 triangle cells by both recurrences, 2080 covering inequalities, 64 extremizer counts, and 354 finite-defect entries with $d \leq 5$. It also matched the first 17 displayed terms of A001192 and the first 25 displayed flattened labeled entries of A182162 [3, 2]. For each $n \leq 7$, the directly enumerated extremal arc polynomial agreed with the Boolean-core construction.

Table 1: Exhaustive enumeration, compared with the recurrence counts. The middle column counts surviving graphs in a fixed topological labeling convention, before isomorphism deduplication.

n	Surviving ordered graphs	Classes u_n	Positive source row $u(n, k)$
1	1	1	1
2	1	1	1
3	2	2	2
4	10	9	8, 1
5	120	88	68, 20
6	3240	1802	1248, 534, 20
7	187920	75598	48640, 25008, 1940, 10

Table 2: Exact unlabeled extremizer counts. Labeled counts are obtained by multiplying the final column by $n!$.

n	q	$a(n)$	$b_n^{(0)}$	n	q	$a(n)$	$b_n^{(0)}$
1	0	1	1	9	4	5	7128
2	1	1	1	10	4	6	8316
3	2	1	2	11	4	7	7128
4	2	2	1	12	4	8	4455
5	3	2	20	13	4	9	1980
6	3	3	20	14	4	10	594
7	3	4	10	15	4	11	108
8	3	5	2	16	4	12	9

The source-count CSV files contain exact integers, not rounded plots or floating-point approximations. A separate `check_asymptotics.py` uses 70-digit Decimal arithmetic to test the first correction at $n = 192, 768, 3072, 12288$ and $d = 0, 1, 2$; all twelve tests improve the leading approximation. Its separate CSV records numerical diagnostics, not interval-certified bounds. The JSON report records the checks and their ranges. No finite experiment is presented as a proof of non-P-recursiveness, the entropy theorem, or a limit distribution.

11.3 Computational complexity and numerical stability

For fixed d , (16) uses only $d + 1$ terms after the total counts through $q + d$ have been tabulated. This is an arithmetic-operation statement, not a bit-complexity claim: the binomial coefficients and the answers themselves have $\Theta_d(n)$ bits in the bulk. Near a complete layer, the exact complement form (12) can be much smaller.

Use integer binomial coefficients for exact evaluation. For large floating-point evaluations, use logarithms and either the covering approximation or the all-orders expansion; do not subtract enormous, nearly equal floating-point source-sieve terms near small parameters. The program caps brute-force enumeration at seven vertices to make the exponential nature of that check explicit rather than silently attempting an infeasible extension.

[write] **Added 3 October 2026, batch 85.** In the repository the two programs ship as `code/verify.py` and `code/check_asymptotics.py`, and their outputs under `data/`. The delivered `data/verification-output.txt` is not shipped: it was a byte copy of `data/verification.json`, since the program prints the JSON report it writes. Both programs write beside themselves (into a `data` directory next to the script, the second without an option), so the report's README gives a rerun on a scratch copy in the delivered layout. The intake ran them so on Python 3.14.4: `verify.py --max-n 64 --brute-n 7` passed in about 7s, `check_asymptotics.py` in about 4s; the three CSV files were reproduced byte for byte, the JSON report up to Windows line endings, and both printed transcripts matched the recorded ones. The 17 terms of A001192 and 25 terms of A182162 that `verify.py` embeds as test fixtures are OEIS data, published under CC BY-SA 4.0.

12 Formalization route and further research

12.1 A finite, auditable formalization route

A formal development need not begin with the full set-theoretic collapse. Represent a finite digraph by a Boolean relation on a finite vertex type. Define the finite set of out-neighbors, sourcehood, acyclicity, and extensionality. The maximum-source proof then reduces to an injection from vertices into the finite powerset of the nonsources, followed by the explicit construction in Theorem 3.1.

A useful next layer would prove that deletion of sources preserves out-neighborhoods and extensionality. This supports a marked-source bijection and finite binomial inversion. To treat unlabeled counts, one may either prove rigidity and use labeled counts throughout, or introduce finite collapse and quotient by graph isomorphism. The former avoids set-theoretic infrastructure for the first main counting results.

The entropy and non-holonomicity proofs can be separated from graph semantics. A generic exponential-jump lemma is particularly suitable for reuse: its hypotheses are scalar asymptotic statements, and its proof only needs eventual polynomial-ratio bounds and the triangle inequality. The deliverable contains no Lean source and makes no claim of kernel verification. This paragraph is a development plan, not an assertion that particular library declarations already implement the argument.

12.2 Specific research questions

The following questions are proposed directions, not claims that each has been certified open in the published literature.

1. Natural boundaries of the extremizer generating functions. Does $B_d(z)$ have a natural boundary on $|z| = e^{-\lambda_{d+1}}$? The proof here establishes non-D-finiteness but not a dense set of singularities. Unlike the lacunary maximum sequence, B_d has positive coefficients throughout its eventual support. A useful first target is an analysis of contributions from short intervals immediately after powers of two.

2. A sharper small-core amplitude. Combine an explicit, verified asymptotic expansion for u_m with (35). Since $m \asymp \log n$, the resulting corrections live on scales different from the Boolean-layer saddle. Distinguish statements uniform in the dyadic phase from fixed-phase expansions.

3. Defects growing with the graph size. How far can the entropy law and exponential-jump obstruction be extended when $d = d(n)$ increases? The covering theorem itself remains valid, but $\log u_{q+d}$ ceases to be a negligible amplitude if the core becomes too large. Identify the transition at which the large-source boundary analysis must connect with unrestricted EAD enumeration.

4. A matched bulk-to-complete-layer expansion. The bulk formula and the fixed-deletion formula overlap when the number $r = 2^m - n$ tends to infinity slowly. Derive explicit uniform remainder bounds covering r from fixed values to a positive fraction of 2^m , without using a Gaussian prefactor at a degenerate endpoint.

5. Extend the arc law beyond the collision estimate. The total-variation coupling loses strength when r is comparable to $2^{m/2}$. This is a limitation of comparing whole subset samples, not necessarily of comparing their sizes. Can a finite-population local limit theorem recover the arc distribution for all densities, with an explicit correction for the m forbidden core subsets?

6. Joint degree statistics. Study the empirical outdegree profile of source vertices and the joint indegree distribution of core vertices. The source-neighborhood family is a sample without replacement from a Boolean cube with a small deleted family. This suggests multivariate hypergeometric limits, but the exact core-induced correlations should be retained rather than discarded.

7. Source layers under repeated deletion. Successively delete all sources. Conditional on the first layer being extremal, the remaining graph is exactly a uniform small core. Can this be iterated to obtain precise distributions for the number of rounds, the full layer-size vector, or the maximum rank under extremal source conditioning?

8. Restricted degree and weak extensionality. If outdegrees are at most Δ , the Boolean capacity becomes $\sum_{j \leq \Delta} \binom{m}{j}$ rather than 2^m . Determine when its pigeonhole bound is sharp, how covering changes, and whether analogous boundary jumps imply non-P-recursiveness. A related problem permits bounded multiplicity of out-neighborhoods or distinguished atoms.

9. Exact ranking and sampling of boundary classes. Implement ranking and unranking for the core-plus-subset representation. A ranked core together with a combinatorial-number-system rank for its chosen neighborhoods gives a natural interface. Specify bit complexity, including the cost of representing deeply nested finite sets.

10. Formal proofs and OEIS editorial updates. Formalize the source support theorem and the boundary bijection first. After independent mathematical review, the proof of A182220 and the boundary formula can be proposed to OEIS with the historical upper-bound citation. No OEIS submission or repository commit was made during this work.

13 Conclusion

The source maximum has a short Boolean-capacity proof, and its known upper bound should not be confused with a new breakthrough. The interesting structure appears when one counts the graphs attaining, or nearly attaining, that maximum. Their core has logarithmic size, while their sources select a large family of subsets of that core. At maximum density, coverage is automatic and the count is exactly $u_q \binom{2^q - q}{n - q}$. At fixed defect, only finitely many terms of the classical sieve survive.

This decomposition makes it possible to prove a uniform entropy law without solving the separate asymptotic problem for the core count. Its phase resets at powers of two. The resulting exponential jumps rule out every polynomial-coefficient linear recurrence for each fixed boundary diagonal. Exact arc formulas and the binomial coupling provide further information about the extremizers themselves.

The principal deliverable is therefore a chain of proved results: sharp source support, exact boundary enumeration, finite-defect formulas, uniform high-source estimates, complete growth-rate intervals, and non-holonomicity, with reproducible finite checks. Questions about historical priority, natural boundaries of the extremizer series, growing defects, and full formal verification remain explicitly separate.

A Suggested concise OEIS-facing statements

A182220. For every $n \geq 1$, $a(n) = n - \lceil \log_2 n \rceil$. Proof: if k vertices are sources, all n distinct out-neighborhoods are subsets of the $n - k$ nonsources, so $n \leq 2^{n-k}$. Conversely, take an extensional acyclic core on $m = \lceil \log_2 n \rceil$ vertices and choose $n - m$ unused subsets as new source neighborhoods. Since $n > 2^{m-1}$ for $m \geq 1$, any resulting family covers the core, so it has exactly $n - m$ sources. The case $n = 1$ is immediate. The upper bound already appears in Tomescu's 2011 thesis, printed page 29.

A182162. Let $T(n, k)$ denote the labeled count. For $q = \lceil \log_2 n \rceil$,

$$T(n, n - q) = n! A001192(q) \binom{2^q - q}{n - q}.$$

More generally, for $d \geq 0$ and $k = n - q - d \geq 1$, formula (16), multiplied by $n!$, gives an exact $(d + 1)$ -term expression. These are specializations of the marked-source sieve, with the extremal case also admitting the direct Boolean-core bijection.

Boundary sequence for a possible separate entry. The unlabeled extremizer sequence begins

1, 1, 2, 1, 20, 20, 10, 2, 7128, 8316, 7128, 4455, 1980, 594, 108, 9, . . .

It satisfies $b_{2^m}^{(0)} = A001192(m)$, $\liminf(b_n^{(0)})^{1/n} = 1$, and $\limsup(b_n^{(0)})^{1/n} = 4$, and it is not P-recursive. No assertion is made that this sequence currently lacks an OEIS entry; a sequence-value and literature search should precede any new-entry proposal.

[write] **Added 3 October 2026, batch 85.** This appendix is draft text. Nothing in it was submitted to OEIS, by the package’s author (question 10 of Section 12.2) or by the intake, and this report submits nothing. Any proposal must keep Tomescu’s credit for the upper bound and is a human editor’s decision after independent review. The intake did not search OEIS for the boundary sequence displayed above.

B Proof-status ledger

Statement	Status and qualification
Maximum-source upper bound	Known in Tomescu (2011); independently proved here.
Maximum formula and full support	Proved constructively here; supplies a proof of the displayed OEIS conjecture. No historical-first claim.
Finite collapse and rigidity	Classical; proofs included to fix semantics and labeling factors.
General source sieve	Existing enumeration; proof included and attribution retained.
Exact extremizer count	Proved by a classification and also by specialization of the sieve; no priority certification.
Finite-defect formula	Exact specialization with a proved $d + 1$ -term truncation.
Uniform covering bound	Proved by a corewise probability estimate.
Entropy law and all limit rates	Proved uniformly for each fixed defect using elementary bounds on u_m .
Non-P-recursiveness	Proved for every fixed defect, including labeled counts.
All-orders expansion	Proved away from $p = 1$, with the core amplitude retained exactly.
Boundary binomial arc law	Proved in total variation with an explicit finite bound.
Numerical verification	Exact recurrence checks through 64 and exhaustive class enumeration through 7; not a substitute for the proofs.
Lean or Rocq verification	Not performed; no such certification claimed.
Natural boundary for B_d	Not proved; proposed as a further question.
Historical priority of refinements	Not established by the available search.

References

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- [2] OEIS Foundation, *A182162: Extensional acyclic digraphs by sources*, <https://oeis.org/A182162>. Includes Nathaniel Johnston’s marked-source-sieve Maple program, April 18, 2012. Consulted October 3, 2026.

- [3] OEIS Foundation, *A001192: Number of full sets of size n* , <https://oeis.org/A001192>. Consulted October 3, 2026.
- [4] OEIS Foundation, *A182161: Number of extensional acyclic digraphs on n labeled nodes*, <https://oeis.org/A182161>. Consulted October 3, 2026.
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- [8] Stephan Wagner, *Asymptotic enumeration of extensional acyclic digraphs*, Proceedings of ANALCO 2012, 1–8. <https://doi.org/10.1137/1.9781611973020.1>. Publisher abstract inspected. The author-hosted full-text link did not respond during this research; no unchecked theorem from it is used in our proofs.
- [9] Vladimir Reshetnikov, *ProveIt*, repository snapshot <https://github.com/VladimirReshetnikov/ProveIt>, commit 6bf7f30d0352f7596e70928b3d4f304914075907, inspected October 3, 2026. Used for project context and a limited nonduplication search, not as a mathematical premise.